

TeaBetes: A Healthcare Analytics Framework for Diabetes Risk Assessment Using Machine Learning, Feature Engineering, and Interactive Dashboards

1st Devysson Souza Cardoso 2nd João Filipe Gouveia Mesquita 3rd Tiago Pinto Silva

Departamento de Ciência e Tecnologia
Universidade Portucalense Infante D. Henrique
Porto, Portugal

48330@alunos.upt.pt 48953@alunos.upt.pt 50440@alunos.upt.pt

4th Thiago Silva Rocha 5th David Filipe dos Santos Aroso

Departamento de Ciência e Tecnologia
Universidade Portucalense Infante D. Henrique
Porto, Portugal

48621@alunos.upt.pt 50441@alunos.upt.pt

Abstract—Diabetes remains one of the most significant global health challenges, motivating the development of data-driven healthcare analytics systems capable of supporting early diagnosis and risk assessment. This work presents a comprehensive analytical pipeline for diabetes-related data analysis combining exploratory data analysis, feature engineering, supervised learning, dimensionality reduction, clustering techniques, and interactive visualization tools.

The study was conducted using a large-scale healthcare dataset containing clinical, behavioral, demographic, and lifestyle-related variables. Multiple classification and regression models were evaluated, including Logistic Regression, Support Vector Machines, Random Forests, boosting methods, neural networks, and ensemble approaches. The classification experiments prioritized recall-oriented optimization due to the clinical importance of minimizing false negatives in diabetes screening scenarios.

The experimental results demonstrated that recall-oriented models such as the Bagging Classifier and Logistic Regression achieved the highest sensitivity, while ensemble methods including AdaBoost, Stacking, and XGBoost provided the best overall balance between predictive performance and computational efficiency. Regression experiments achieved extremely high predictive performance, with the Stacking Regressor emerging as the strongest model evaluated on the complete dataset, characterizing the problem as a clinical risk reconstruction task strongly associated with the available predictors.

Additionally, PCA and clustering techniques were applied to identify latent patient subgroup structures and evaluate the effects of dimensionality reduction. The results showed that PCA improved clustering stability while negatively affecting tree-based regression models. Finally, an interactive Streamlit dashboard was developed to support exploratory healthcare analytics and visualization workflows for both population-level and individual-level analysis.

Overall, the results demonstrate the effectiveness of combining feature engineering, machine learning, dimensionality reduction, clustering, and interactive healthcare analytics for diabetes-related studies. Furthermore, the developed Streamlit platform operationalizes the selected champion models through real-time

predictive inference, enabling both diabetes diagnosis prediction and continuous risk score estimation within an interactive healthcare analytics environment.

I. INTRODUCTION

Diabetes is one of the most prevalent chronic diseases worldwide and represents a major public health challenge. According to recent global health reports, the number of individuals affected by diabetes continues to increase due to factors such as sedentary lifestyles, poor dietary habits, obesity, and population aging. If not detected and managed early, diabetes may lead to severe complications including cardiovascular diseases, kidney failure, neuropathy, and vision loss.

The growing availability of healthcare data has enabled the application of data-driven approaches to support diabetes analysis, prediction, and risk assessment. In recent years, machine learning techniques have become increasingly important in healthcare analytics due to their ability to identify complex patterns and relationships within high-dimensional clinical and behavioral datasets. These approaches can support healthcare professionals by improving screening processes, identifying high-risk patients, and assisting exploratory medical analysis.

In addition to predictive modeling, exploratory data analysis and feature engineering play fundamental roles in healthcare machine learning pipelines. Clinical variables such as glucose levels, insulin measurements, body mass index (BMI), blood pressure, and cholesterol indicators are strongly associated with diabetes risk. Furthermore, behavioral and lifestyle-related factors, including physical activity, diet quality, sleep duration, smoking habits, and alcohol consumption, also contribute significantly to metabolic health outcomes.

Recent advances in healthcare analytics increasingly combine supervised learning, dimensionality reduction, clustering techniques, and interactive visualization systems to improve both predictive performance and interpretability. Ensemble learning methods, neural networks, and explainable artificial intelligence approaches have demonstrated promising results in diabetes-related prediction tasks. Simultaneously, interactive dashboards and visual analytics systems are becoming valuable tools for supporting exploratory healthcare analysis and improving accessibility to complex analytical results.

This work presents a comprehensive healthcare analytics pipeline for diabetes-related data analysis using a large-scale structured dataset containing clinical, demographic, behavioral, and lifestyle-related variables. The proposed methodology integrates exploratory data analysis, feature engineering, supervised learning models, hyperparameter optimization, dimensionality reduction, clustering techniques, and interactive dashboard visualization.

The classification experiments evaluate multiple tradeoffs between recall, overall predictive performance, computational efficiency, and interpretability. Particular attention is given to recall-oriented models due to the clinical importance of minimizing false negatives in diabetes screening scenarios. Additionally, regression models are employed to reconstruct a diabetes risk score derived from the available clinical and behavioral indicators. Unsupervised learning techniques are also applied to explore latent subgroup patterns within the dataset.

Finally, a Streamlit-based dashboard is developed to support interactive healthcare analytics from both clinical and individual perspectives. The platform is organized into two complementary environments: a patient-oriented predictive interface and a doctor-oriented analytical interface. The patient dashboard integrates the machine learning models selected for deployment following the experimental evaluation phase, enabling real-time diabetes diagnosis prediction and continuous diabetes risk score estimation. In parallel, the physician dashboard focuses on population-level exploratory analytics, providing interactive visualization, cohort exploration, and clinical pattern analysis capabilities. Together, these components combine predictive inference with healthcare analytics in a unified decision-support environment.

Overall, this work aims to demonstrate how the integration of machine learning, engineered healthcare features, dimensionality reduction, clustering analysis, and interactive visualization can support robust and interpretable diabetes-related healthcare analytics systems.

II. RELATED WORK

Diabetes remains one of the most significant global health challenges, motivating extensive research in data analysis and machine learning for early diagnosis and risk prediction. In recent years, the increasing availability of healthcare datasets has enabled the development of predictive models capable of identifying diabetes risk patterns using clinical, behavioral, and socio-economic variables [1], [2].

Recent studies have explored the application of machine learning techniques to structured healthcare datasets containing variables such as glucose levels, blood pressure, insulin measurements, body mass index (BMI), cholesterol levels, physical activity, and lifestyle habits [3], [4]. Research demonstrates that combining clinical and behavioral data improves model robustness and predictive performance, since diabetes is influenced by both physiological and lifestyle-related factors.

Several machine learning algorithms have been widely employed in diabetes prediction tasks. Traditional models such as Logistic Regression and Naïve Bayes continue to be used due to their simplicity and interpretability. However, recent works show that more advanced approaches, including Random Forest, Support Vector Machines (SVM), Gradient Boosting, XGBoost, and Neural Networks, generally achieve superior accuracy by capturing complex and non-linear relationships within the data [2], [5].

Recent research trends also emphasize the use of ensemble learning techniques. Studies published in recent years report that ensemble models such as Random Forest and XGBoost outperform individual classifiers in diabetes prediction tasks due to their ability to reduce overfitting and improve generalization [5], [6]. These methods are particularly effective when dealing with heterogeneous healthcare datasets that combine numerical, categorical, and behavioral variables.

Another important trend in recent literature is the adoption of Explainable Artificial Intelligence (XAI) techniques in healthcare analytics [4], [7]. Since predictive models in medical contexts require transparency and interpretability, several studies use SHAP (SHapley Additive Explanations) and feature importance analysis to identify the variables that most influence diabetes prediction. Research consistently identifies features such as fasting glucose, HbA1c, BMI, age, physical activity, and family history as highly relevant predictors.

Exploratory Data Analysis (EDA) also plays a central role in healthcare machine learning pipelines, supporting feature selection and data understanding [2], [8]. Recent works emphasize the importance of correlation analysis, distribution analysis, outlier detection, and dimensionality reduction techniques before applying machine learning models. EDA enables researchers to better understand data patterns, identify redundant features, and support feature selection strategies, contributing to improved model performance and interpretability.

A. Feature Engineering in Diabetes Prediction

Feature engineering has become a major focus in recent diabetes analytics research, improving both model performance and interpretability [3], [5]. Instead of relying exclusively on raw clinical variables, recent studies propose the creation of derived indicators capable of better representing physiological and metabolic relationships.

One widely explored area is the estimation of insulin resistance using combinations of fasting glucose and insulin measurements. Recent healthcare analytics studies show that composite indicators related to insulin sensitivity improve

predictive accuracy and provide better representations of metabolic dysfunction [3].

Similarly, glycemic variability has gained significant attention in recent years. Research indicates that the difference between fasting and postprandial glucose levels can reveal early metabolic abnormalities associated with diabetes progression [4]. Derived variables that capture glucose fluctuations are increasingly used to improve predictive performance.

Recent studies also emphasize the importance of obesity-related composite indicators. Instead of analyzing BMI independently, modern approaches combine BMI with other anthropometric measures to better estimate metabolic risk and cardiovascular complications [1], [2].

Cardiovascular indicators derived from blood pressure variables, such as pulse pressure and mean arterial pressure (MAP), are also commonly used in healthcare analytics pipelines. These variables provide more comprehensive representations of vascular health and are associated with long-term diabetes complications [3].

In addition, recent literature highlights the relevance of lipid profile ratios and aggregated clinical risk scores. Composite indicators combining clinical history, lifestyle variables, and cardiovascular measurements improve both predictive capability and model interpretability [6], [7].

Current research trends demonstrate that integrating clinical data, behavioral information, exploratory data analysis, and engineered features leads to more accurate and explainable diabetes prediction systems. The methodology adopted in this work follows these recent approaches by combining raw healthcare variables with engineered features designed to capture metabolic, cardiovascular, and behavioral risk factors.

In healthcare screening applications, evaluation metrics must be interpreted according to clinical priorities. Several recent studies emphasize that recall-oriented models are particularly important in disease screening tasks, since false negatives may delay diagnosis and treatment. Consequently, many healthcare machine learning systems prioritize sensitivity and recall over overall accuracy when early detection is the primary objective [4], [7].

III. METHODOLOGY

This work follows a data-driven healthcare analytics methodology combining exploratory data analysis, feature engineering, supervised learning, unsupervised learning, dimensionality reduction, and interactive visualization techniques. The objective is to analyze diabetes-related risk patterns using clinical, behavioral, and socio-economic variables extracted from a large-scale structured dataset.

A. Dataset Description

The experiments were conducted using the dataset `diabetes_dataset_new_variables.csv`, containing approximately 100,000 patient records and 40 original attributes. The dataset includes demographic, physiological, behavioral, and lifestyle-related variables associated with diabetes risk.

The variables can be grouped into the following categories:

- **Clinical variables:** fasting glucose, postprandial glucose, HbA1c, insulin level, systolic blood pressure, cholesterol indicators, pulse pressure, and mean arterial pressure.
- **Behavioral and lifestyle variables:** physical activity, diet quality, sleep duration, alcohol consumption, smoking habits, and screen time.
- **Demographic variables:** age, gender, ethnicity, education level, employment status, and income level.
- **Derived variables:** insulin resistance indicators, obesity-related indexes, cardiovascular risk indicators, and composite diabetes risk scores.

Two supervised learning targets were defined:

- `diagnosed_diabetes` for binary classification tasks;
- `diabetes_risk_score` for regression tasks.

For classification experiments, the variables `diagnosed_diabetes` and `diabetes_stage` were excluded from the feature matrix to avoid target leakage. For regression experiments, `diabetes_risk_score`, `diagnosed_diabetes`, and `diabetes_stage` were excluded from the predictors.

The distribution of diabetes stages within the dataset is presented in Fig. 1. The dataset contains a heterogeneous population structure, including non-diabetic, prediabetic, and diagnosed diabetic patients, supporting both supervised and exploratory analytical tasks.

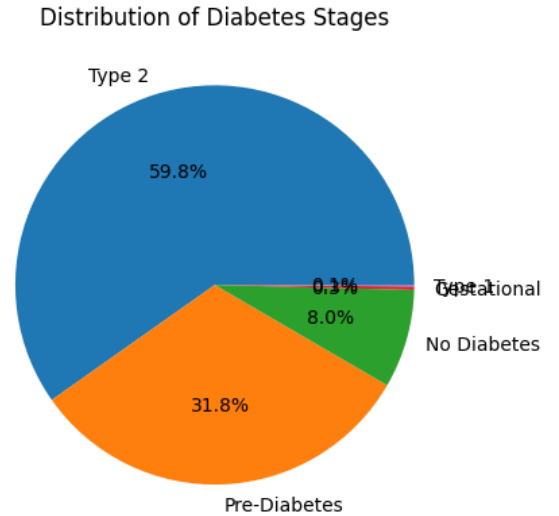


Fig. 1: Distribution of diabetes stages in the analyzed population.

B. Dataset Provenance and Limitations

The dataset used in this work was obtained from a publicly available healthcare analytics repository on Kaggle. The dataset contains approximately 100,000 observations and integrates demographic, clinical, behavioral, and lifestyle-related variables associated with diabetes risk.

Although the dataset provides a rich environment for exploratory analytics and machine learning experimentation, some limitations must be considered. The dataset documentation does not provide detailed information regarding data collection procedures, population sampling methodology, or clinical acquisition protocols. Additionally, some variables and risk indicators appear to follow deterministic or semi-simulated relationships commonly observed in educational healthcare datasets.

Consequently, the results presented in this work should be interpreted primarily within the context of healthcare analytics experimentation and methodological evaluation rather than direct clinical deployment. Despite these limitations, the dataset remains suitable for evaluating machine learning pipelines, feature engineering strategies, exploratory analytics workflows, and visualization systems in diabetes-related studies.

C. Data Preprocessing

Categorical variables were transformed using one-hot encoding with the `drop_first=True` strategy to reduce multicollinearity. After encoding, the final feature matrix contained 53 predictive variables.

Different preprocessing pipelines were adopted according to the characteristics of each machine learning model:

- Tree-based models such as Decision Trees, Random Forests, Gradient Boosting, and XGBoost were trained without feature scaling, since these algorithms are scale-invariant.
- Distance-based and gradient-based models, including K-Nearest Neighbors (KNN), Support Vector Machines (SVM/SVR), Perceptron, and Multi-Layer Perceptrons (MLP), used either standardization or normalization.
- Standardization was implemented using `StandardScaler()`, while normalization-based pipelines were applied in selected linear and support-vector experiments.

The dataset was split into training and testing subsets using an 80/20 ratio with stratification for classification tasks and a fixed random state of 42 to ensure reproducibility.

D. Feature Engineering

Feature engineering was applied to create clinically meaningful indicators capable of better representing physiological and behavioral relationships associated with diabetes risk.

Several composite variables were generated, including:

- **Insulin resistance indicators**, combining glucose and insulin measurements;
- **Glycemic variability features**, derived from fasting and postprandial glucose levels;
- **Obesity-related indicators**, combining BMI and body composition variables;
- **Cardiovascular indicators**, including pulse pressure and mean arterial pressure (MAP);
- **Lipid profile ratios**, such as LDL/HDL cholesterol ratio;

- **Lifestyle composite scores**, integrating physical activity, diet quality, sleep duration, and sedentary behavior.

Some engineered variables were derived from established healthcare indicators and composite formulations commonly used in cardiovascular and metabolic analysis. For example, mean arterial pressure (MAP) was computed as:

$$MAP = \frac{SBP + 2 \times DBP}{3} \quad (1)$$

where *SBP* represents systolic blood pressure and *DBP* represents diastolic blood pressure.

Similarly, pulse pressure was calculated as:

$$PP = SBP - DBP \quad (2)$$

Additional engineered features combined glucose, insulin, BMI, cholesterol, and lifestyle-related variables to better represent metabolic and behavioral risk patterns associated with diabetes.

These engineered features were designed to improve both predictive performance and clinical interpretability.

Fig. 2 illustrates the distribution of the engineered insulin resistance indicator generated during the feature engineering stage. The distribution demonstrates substantial variability across the population, reinforcing its usefulness as a discriminative metabolic feature.

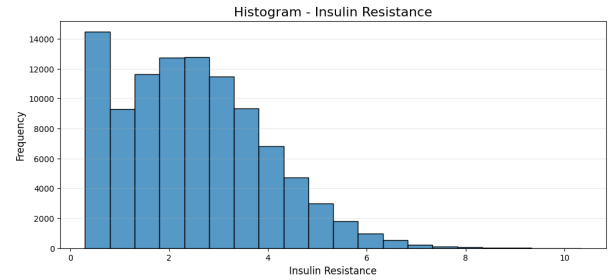


Fig. 2: Distribution of the engineered insulin resistance index.

E. Supervised Learning Models

A broad set of classification and regression algorithms was evaluated in order to compare predictive performance, computational cost, and interpretability.

1) *Classification Models*: The following classification algorithms were implemented:

- Logistic Regression
- Naïve Bayes
- K-Nearest Neighbors (KNN)
- Support Vector Classifier (SVC)
- Decision Tree Classifier
- Random Forest Classifier
- Gradient Boosting Classifier
- AdaBoost Classifier
- XGBoost Classifier
- Perceptron
- Stacking Classifier

- Bagging Logistic Regression

The classification experiments prioritized recall-oriented optimization due to the clinical importance of minimizing false negatives in diabetes screening tasks.

The evaluated metrics included:

- Accuracy
- Recall
- F1-score
- Training and testing performance
- Computational fit time

2) *Regression Models*: The following regression models were evaluated for diabetes risk score reconstruction:

- Linear Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- AdaBoost Regressor
- XGBoost Regressor
- Support Vector Regressor (SVR)
- KNN Regressor
- MLP Regressor
- Stacking Regressor

Regression performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)
- Computational fit time

F. Hyperparameter Optimization

Hyperparameter optimization was performed using both GridSearchCV and Optuna.

For classification tasks, optimization focused primarily on maximizing recall, while regression optimization minimized MAE. Cross-validation strategies varied according to computational complexity and model characteristics, using either 3-fold or 5-fold validation.

Optuna was employed for efficient hyperparameter search in computationally expensive models, while GridSearchCV was used for exhaustive search in more constrained parameter spaces.

G. Cross Validation

Cross-validation was applied throughout the experiments to improve robustness and reduce overfitting risks.

Classification models generally used stratified cross-validation with recall-oriented scoring, while regression models adopted K-Fold validation with shuffled splits and negative MAE scoring.

This validation strategy enabled reliable comparisons between models and improved the consistency of the experimental results.

H. Dimensionality Reduction

Principal Component Analysis (PCA) was applied in unsupervised learning experiments and dimensionality reduction analysis.

Before PCA, all features were standardized using `StandardScaler()`. The transformation was configured to preserve 95% of the explained variance, which successfully compressed the original feature space from 53 variables down to 27 principal components. This reduced 27-dimensional space was utilized to stabilize the K-Means clustering and was also systematically evaluated across four distinct supervised learning models (Logistic Regression, XGBoost, Ridge, and Random Forest) to assess the mathematical tradeoffs between computational efficiency and predictive performance.

I. Clustering Analysis

Unsupervised learning experiments were performed to explore latent subgroup patterns within the dataset.

K-Means clustering was applied after PCA transformation, with the number of clusters selected using the elbow method. The final configuration adopted $K = 4$, balancing inertia reduction and cluster interpretability.

Hierarchical clustering using Ward linkage was also performed on a random sample of 2,000 observations to validate the K-Means segmentation through dendrogram analysis.

These clustering techniques enabled the exploration of latent subgroup patterns and population structures within the dataset.

J. Dashboard Architecture

An interactive analytical dashboard was developed using Streamlit to support healthcare analytics and decision support. The platform follows a separation-of-concerns architecture composed of two complementary analytical environments: a patient-oriented predictive interface and a doctor-oriented analytical interface.

The dashboard contains two analytical perspectives:

- **Doctor dashboard**: population-level analytics with filtering systems, KPI monitoring, metabolic visualizations, correlation analysis, cohort exploration, and interactive visualization tools.
- **Patient dashboard**: individual-oriented predictive analytics with real-time diabetes diagnosis prediction, continuous diabetes risk estimation, lifestyle indicators, and personalized visual feedback.

The patient dashboard integrates the trained machine learning models developed during this study. User-provided clinical and lifestyle variables are processed through the same preprocessing pipeline used during training, including feature alignment and scaling operations.

Two predictive models are deployed for real-time inference:

- A Bagging Classifier responsible for diabetes diagnosis prediction;
- A Stacking Regressor ensemble responsible for estimating the continuous diabetes risk score.

The deployed models were selected not exclusively according to isolated benchmark rankings, but according to

a combination of predictive performance, inference stability, computational efficiency, and suitability for interactive deployment. The Bagging Classifier was selected due to its strong recall performance and robustness in healthcare screening scenarios, where minimizing false negatives is a primary objective. Similarly, the Stacking Regressor offered a strong balance between predictive accuracy, generalization capability, and robustness across heterogeneous patient profiles, making it suitable for real-time diabetes risk estimation.

This architecture allows users to interactively modify health-related variables and immediately observe changes in their predicted risk profile, transforming the dashboard into an interactive healthcare simulation and awareness tool.

To operationalize these models, a dedicated preprocessing module was implemented in the application backend. This component receives individual patient inputs, applies the same preprocessing transformations used during model training, and dynamically aligns categorical features through one-hot encoding. Missing categorical dimensions are automatically padded with zero values to preserve compatibility with the trained feature space. In addition, the deployed models are loaded together with their corresponding preprocessing objects, ensuring that scaling and feature transformations remain consistent between training and inference environments. This approach guarantees reproducible predictions and prevents training-serving skew during real-time execution.

Conversely, the doctor dashboard was intentionally designed as a business intelligence and exploratory analytics environment. Rather than performing large-scale batch inference, it focuses on population-level analysis through interactive filtering, cohort exploration, KPI monitoring, correlation analysis, and clinical pattern discovery. This architectural separation improves scalability, preserves responsiveness, and aligns each interface with its intended analytical objective.

IV. RESULTS AND DISCUSSION

This section presents the main findings obtained from the supervised learning, dimensionality reduction, clustering, and dashboard experiments. The results are analyzed considering predictive performance, computational cost, interpretability, and clinical applicability.

A. Classification Results

Table I summarizes the best-performing classification models evaluated in this work.

The tradeoff between recall-oriented optimization and overall predictive balance is illustrated in Fig. 4. Models emphasizing sensitivity generally achieved higher recall values at the expense of overall balance, while ensemble methods provided more consistent performance across all evaluation metrics.

Although the dataset does not present extreme class imbalance, recall-oriented evaluation remains particularly important in healthcare screening scenarios due to the clinical consequences associated with false negatives. In diabetes screening applications, failing to identify high-risk patients may delay diagnosis and treatment, making sensitivity-oriented evaluation strategies clinically relevant.

The classification experiments revealed important tradeoffs between recall, overall accuracy, computational efficiency, and interpretability.

The Bagging Classifier achieved the highest recall value (0.9202), making it the most clinically sensitive model among the final evaluated approaches. Logistic Regression produced a very similar recall value (0.9156), reinforcing the effectiveness of linear decision boundaries for diabetes screening tasks. Since minimizing false negatives is a primary objective in healthcare screening applications, these results are particularly relevant from a clinical perspective.

The classification error distribution of the deployed Bagging Classifier is illustrated in Fig. 5. The confusion matrix highlights the model's strong ability to identify positive diabetes cases while also demonstrating the increased number of false positives associated with recall-oriented optimization strategies.

While Bagging and Logistic Regression prioritized recall, ensemble approaches such as AdaBoost, Gradient Boosting, and Stacking achieved the best overall balance between accuracy and F1-score. AdaBoost obtained the highest accuracy (0.9199) and F1-score (0.9285), demonstrating the effectiveness of boosting techniques when combining heterogeneous clinical and behavioral variables.

XGBoost demonstrated one of the most interesting results in the study. Although its recall was slightly lower than the recall-oriented models, it achieved excellent accuracy and F1-score while presenting an extremely low computational cost. The model required only 0.45 seconds for training, making it the most computationally efficient high-performance classifier evaluated.

The Stacking model achieved competitive performance across all evaluation metrics and was selected as a representative ensemble approach due to its robustness and stable behavior. However, the deployed classification pipeline prioritized the Bagging Classifier because of its superior recall performance and suitability for healthcare screening environments.

The SVC model demonstrated strong generalization capabilities but presented extremely high computational costs, even after reducing the training dataset size. This highlights one of the major limitations of kernel-based methods in large-scale healthcare datasets.

“ “

B. Regression Results

Fig. 6 summarizes the regression reconstruction performance across the evaluated models. Neural and ensemble-based approaches achieved substantially lower reconstruction errors compared to distance-based and single-tree models.

The regression experiments focused on reconstructing the `diabetes_risk_score` target using clinical and behavioral predictors.

To ensure fair comparison and consistent evaluation conditions, only models successfully trained and evaluated on the complete 100,000-row dataset were included in the final regression ranking. Although additional approaches such as

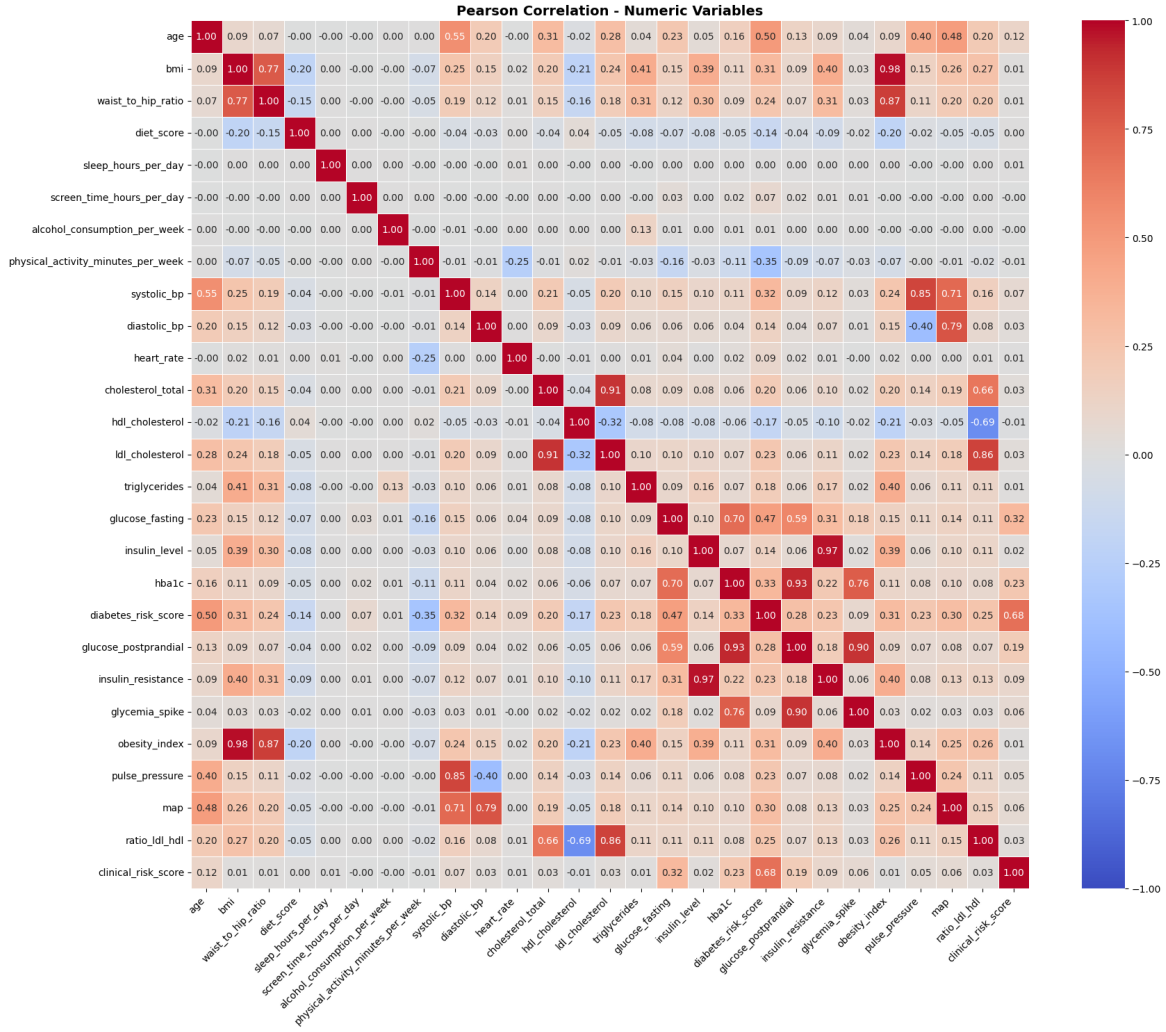


Fig. 3: Pearson correlation matrix for the main numerical diabetes-related variables.

TABLE I: Champion Classification Results

Algorithm	Accuracy	Recall	F1	Fit Time (s)	Main Characteristic
Decision Tree	0.9174	0.8683	0.9266	0.67	Fast interpretable model
Naive Bayes	0.8237	0.8263	0.8490	24.94	Probabilistic baseline
Logistic Regression	0.8116	0.9156	0.8536	17.52	Strong recall baseline
Perceptron	0.8937	0.8228	0.9028	8.79	Linear classifier baseline
Bagging Classifier	0.8050	0.9202	0.8499	23.49	Deployed recall ensemble
Random Forest	0.9168	0.8686	0.9260	117.12	Balanced ensemble
AdaBoost	0.9199	0.8665	0.9285	6.68	Best overall balance
Gradient Boosting	0.9199	0.8665	0.9284	27.41	Strong boosting model
XGBoost	0.9176	0.8674	0.9266	0.45	Best efficiency ratio
Stacking	0.9197	0.8668	0.9283	153.90	Deployed ensemble model

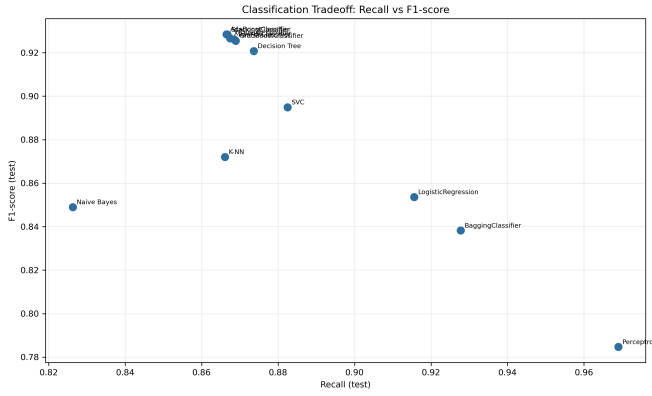


Fig. 4: Recall versus F1-score for the champion classification run of each model family.

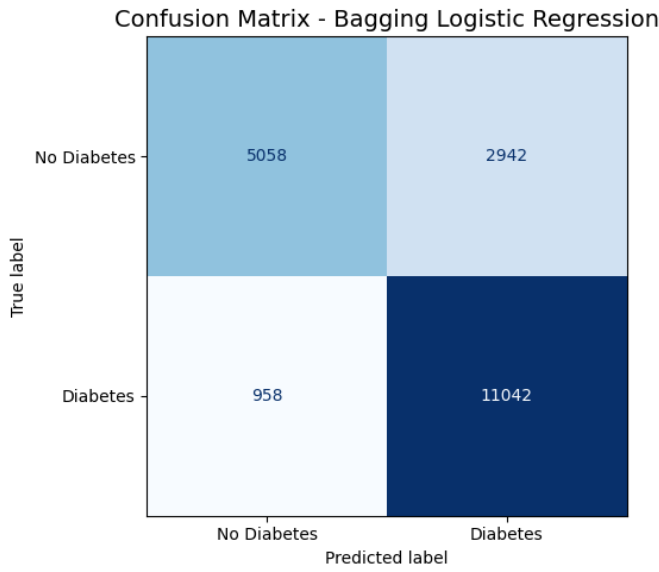


Fig. 5: Confusion matrix of the deployed Bagging Classifier highlighting the tradeoff between sensitivity and false-positive generation.

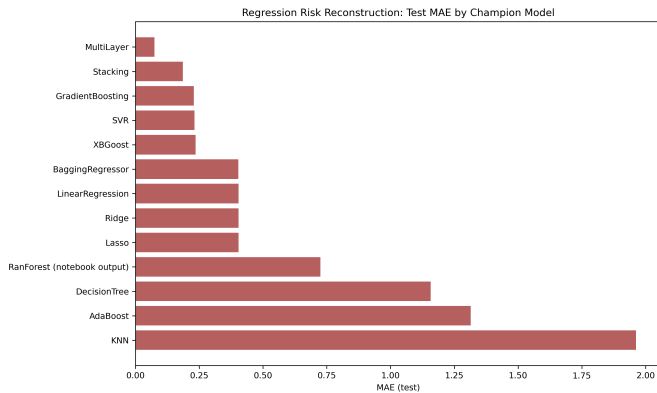


Fig. 6: Test MAE comparison for champion regression models in diabetes risk score reconstruction.

MLP Regressor and Support Vector Regression (SVR) were explored during experimentation, their computational requirements necessitated execution on a reduced 50,000-row subset. Consequently, these models were excluded from the final comparison presented in Table II.

Table II summarizes the best regression results.

The Stacking Regressor achieved the best overall regression performance among the models evaluated on the complete dataset, obtaining an R^2 value above 0.999 and extremely low reconstruction errors. Gradient Boosting and XGBoost also demonstrated excellent predictive capabilities, indicating that the diabetes risk score can be reconstructed with very high precision from the available clinical and behavioral variables.

However, the regression results must be interpreted carefully. The target variable, `diabetes_risk_score`, is strongly derived from the same clinical and behavioral variables used as predictors. Therefore, the regression task should be interpreted as a clinical risk reconstruction problem rather than a purely latent prediction problem.

This interpretation is supported by the extremely strong performance obtained even by simple linear models. Ordinary Least Squares (OLS), Ridge, Lasso, and Bagging Regression all achieved R^2 values close to 0.994, indicating strong deterministic relationships between predictors and the target construction process.

The Stacking Regressor combined multiple learning paradigms into a single ensemble architecture and consistently produced the most accurate reconstruction results among the models evaluated on the full dataset. Its strong performance, combined with robust generalization characteristics and stable behavior across heterogeneous patient profiles, motivated its deployment within the Streamlit platform for real-time diabetes risk estimation.

Linear Regression emerged as the fastest regression approach while still maintaining excellent predictive capability, reinforcing the deterministic nature of the regression target. Although additional models such as MLP Regressor and SVR achieved competitive results in preliminary experiments, they were excluded from the final comparison because they were evaluated only on a reduced 50,000-row subset rather than the complete dataset.

C. Hyperparameter Optimization Analysis

Both GridSearchCV and Optuna were employed throughout the experiments for hyperparameter optimization.

Optuna demonstrated strong performance in large search spaces and computationally expensive models, particularly for Perceptron, SVC, MLP, and boosting approaches. In several cases, Optuna improved recall-oriented classification performance and reduced regression error metrics.

However, GridSearchCV remained competitive for constrained parameter spaces, especially in interpretable models such as Decision Trees and KNN. This suggests that exhaustive search can still be effective when the parameter space is relatively small and well-structured.

TABLE II: Champion Regression Results

Model	MAE	RMSE	R^2	Fit Time (s)	Main Characteristic
Stacking Regressor	0.1859	0.2451	0.9992	252.55	Champion deployed ensemble
Gradient Boosting Regressor	0.2286	0.2951	0.9989	698.99	Strong boosting performance
XGBoost Regressor	0.2357	0.3053	0.9988	42.61	Efficient boosting model
Linear Regression	0.4038	0.7112	0.9938	0.21	Fastest interpretable model
Lasso Regression	0.4038	0.7112	0.9938	46.46	Sparse feature selection
Ridge Regression	0.4038	0.7112	0.9938	28.29	Regularized linear model
Bagging Regressor	0.4032	0.7113	0.9938	—	Ensemble baseline
Random Forest Regressor	0.7249	0.9516	0.9890	—	Balanced tree ensemble
Decision Tree Regressor	1.1578	1.4703	0.9738	50.68	Interpretable tree model
AdaBoost Regressor	1.3146	1.6852	0.9655	—	Weak boosting performance

MLP Regressor and SVR were excluded from the final comparison because they were evaluated only on a 50,000-row subset.

The optimization experiments also revealed that some default or minimally tuned models, particularly XGBoost and AdaBoost, already achieved highly competitive performance. This demonstrates the robustness of modern ensemble methods in structured healthcare datasets.

D. PCA and SVD Analysis

To systematically evaluate the impact of dimensionality reduction, the PCA/SVD representation (retaining 95% variance via 27 components) was tested across linear and tree-based models for both classification and regression.

For linear algorithms, PCA provided substantial computational gains. In the classification task, Logistic Regression fit time decreased by 40.4% (from 0.758s to 0.452s) with only a marginal reduction in Recall (0.8949 to 0.8882). Similarly, in the regression task, Ridge Regression fit time dropped by 67.9% (0.065s to 0.021s), although MAE increased from 0.4039 to 0.4746. These results confirm that orthogonal feature compression accelerates linear optimization by resolving multicollinearity and reducing matrix dimensionality, albeit at a slight cost to predictive precision.

Conversely, PCA negatively impacted tree-based ensembles. For classification, XGBoost fit time slightly increased (+2.2%) while Recall dropped to 0.8552. The degradation was most severe for the Random Forest Regressor (as illustrated in Fig. 7), where fit time increased by 93.5% and MAE more than doubled (from 0.7249 to 1.5095). Tree-based algorithms rely on orthogonal, axis-aligned splits based on original clinical thresholds. Feeding them dense linear combinations destroys feature interpretability and splitting efficiency.

These experiments demonstrate that while PCA is highly effective for clustering tasks and accelerating linear solvers, it is detrimental to tree-based ensembles in this healthcare dataset.

E. Clustering Analysis

Unsupervised clustering experiments suggested the existence of four major exploratory subgroup patterns within the dataset.

The K-Means configuration process is illustrated in Fig. 8, where the elbow criterion suggests that four clusters provide the best tradeoff between inertia reduction and segmentation simplicity.

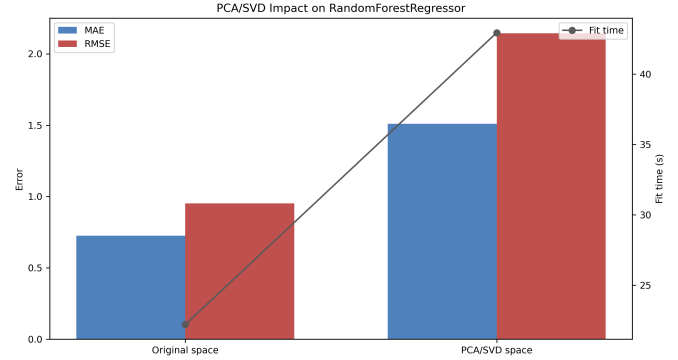


Fig. 7: Performance and training time trade-offs under PCA/SVD dimensionality reduction for tree-based ensemble regression.

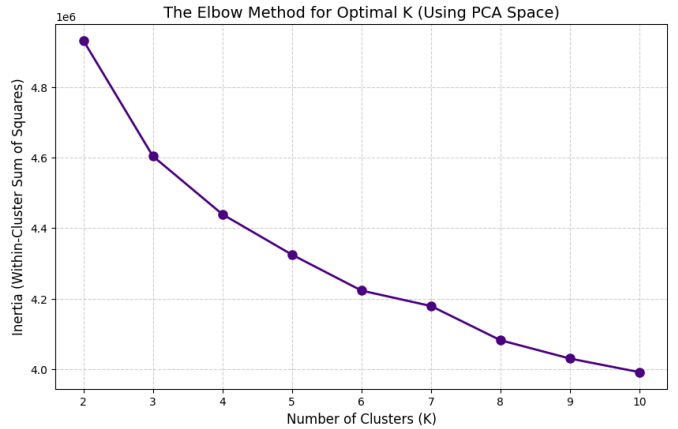


Fig. 8: Elbow method curve for selecting the number of K-Means clusters in PCA space.

The elbow method suggested that $K = 4$ provided the best balance between inertia reduction and cluster interpretability.

The hierarchical clustering structure shown in Fig. 9 further supports the existence of four dominant exploratory subgroup patterns within the dataset.

This result was further explored using hierarchical clustering with Ward linkage, where the dendrogram structure visually supported the existence of four major branches.

The resulting cluster distribution was highly asymmetric,

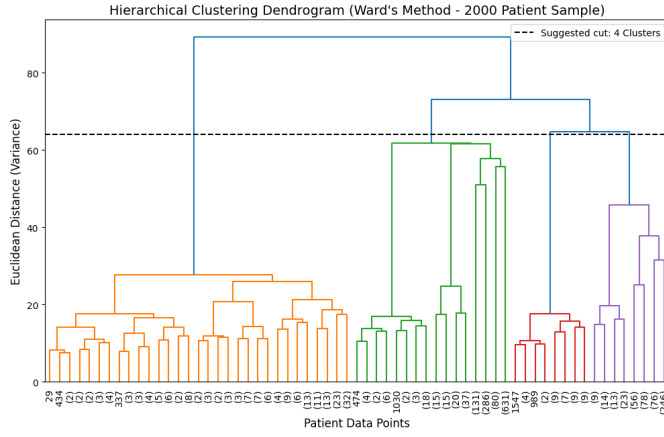


Fig. 9: Ward-linkage hierarchical clustering dendrogram suggesting exploratory subgroup structures within the dataset.

with one dominant cluster containing the majority of observations and smaller clusters representing more specific patient subgroups. This behavior is expected in healthcare datasets, where a large proportion of individuals tend to share relatively similar metabolic and lifestyle characteristics, while smaller groups capture atypical profiles or combinations of risk factors.

Although these clusters should not be interpreted as clinically validated patient categories, the results suggest the presence of meaningful latent structures within the dataset. Such subgroup identification may support future population segmentation strategies, personalized intervention systems, and recommender-system approaches based on patient similarity. Therefore, the clustering results should be viewed as exploratory evidence of heterogeneity in the diabetes population rather than definitive clinical stratification.

F. Dashboard Analysis

The doctor-oriented analytical interface is presented in Fig. 10. The dashboard integrates KPI monitoring, interactive filtering systems, metabolic risk visualization, and exploratory cohort analysis tools designed to support population-level healthcare analytics.

Fig. 11 presents the patient-oriented exploratory interface, which contextualizes individual clinical and lifestyle indicators through personalized visual feedback and machine-learning-based risk estimation mechanisms.

Unlike the physician dashboard, the patient interface integrates the machine learning models selected during the experimental phase. The diagnostic classification is generated by the deployed Bagging Classifier, while the diabetes risk score is estimated through the Stacking Regressor ensemble. This integration enables real-time predictive feedback and allows users to explore how clinical and lifestyle changes may influence their predicted diabetes risk profile.

The developed Streamlit dashboard successfully integrated exploratory healthcare analytics and interactive visualization techniques.

The doctor-oriented dashboard provided population-level analytical tools, including filtering systems, KPI monitoring, metabolic risk visualizations, correlation heatmaps, and exploratory variable analysis tools. These functionalities support clinical hypothesis exploration and descriptive analytics workflows.

The patient-oriented dashboard focused on individualized predictive analytics through machine-learning-based diagnosis prediction, continuous diabetes risk estimation, lifestyle indicators, and personalized behavioral visualizations. The interface contextualized personal metrics against population-level distributions and clinical thresholds while providing real-time predictive feedback generated by the deployed models.

One important aspect of the dashboard architecture is the explicit separation between exploratory analytics and predictive inference. The patient-oriented interface integrates the trained machine learning models developed during this study. Real-time predictions are generated through a Bagging Classifier for diabetes diagnosis and a Stacking Regressor for diabetes risk score estimation. This integration transforms the dashboard into an interactive health simulation environment where users can evaluate the potential impact of behavioral and clinical changes on predicted outcomes.

Conversely, the doctor-oriented dashboard deliberately avoids large-scale predictive inference. Since the platform was designed for population-level analysis involving thousands of records, executing complex ensemble models in real time could introduce substantial computational overhead. Instead, the physician interface focuses on exploratory analytics, cohort segmentation, KPI monitoring, and clinical pattern discovery. This architectural separation improves scalability, preserves responsiveness, and aligns each interface with its intended analytical objective.

Overall, the dashboard complements the machine learning pipeline by transforming analytical findings into interactive healthcare visualization experiences suitable for both clinical and individual perspectives.

G. Limitations

Despite the strong analytical performance obtained throughout the experiments, several limitations must be acknowledged.

First, the dataset used in this study originates from a public healthcare analytics repository and does not provide detailed information regarding clinical acquisition procedures, population sampling strategies, demographic representativeness, or data collection protocols. Consequently, the results should not be interpreted as clinically validated diagnostic performance, but rather as healthcare analytics outcomes evaluated within the scope of the available dataset.

Second, the regression task involves reconstructing a diabetes risk score that is strongly associated with the predictor variables themselves. Therefore, the extremely high regression performance primarily reflects deterministic or semi-deterministic relationships embedded within the dataset structure. While this makes the task useful for methodological

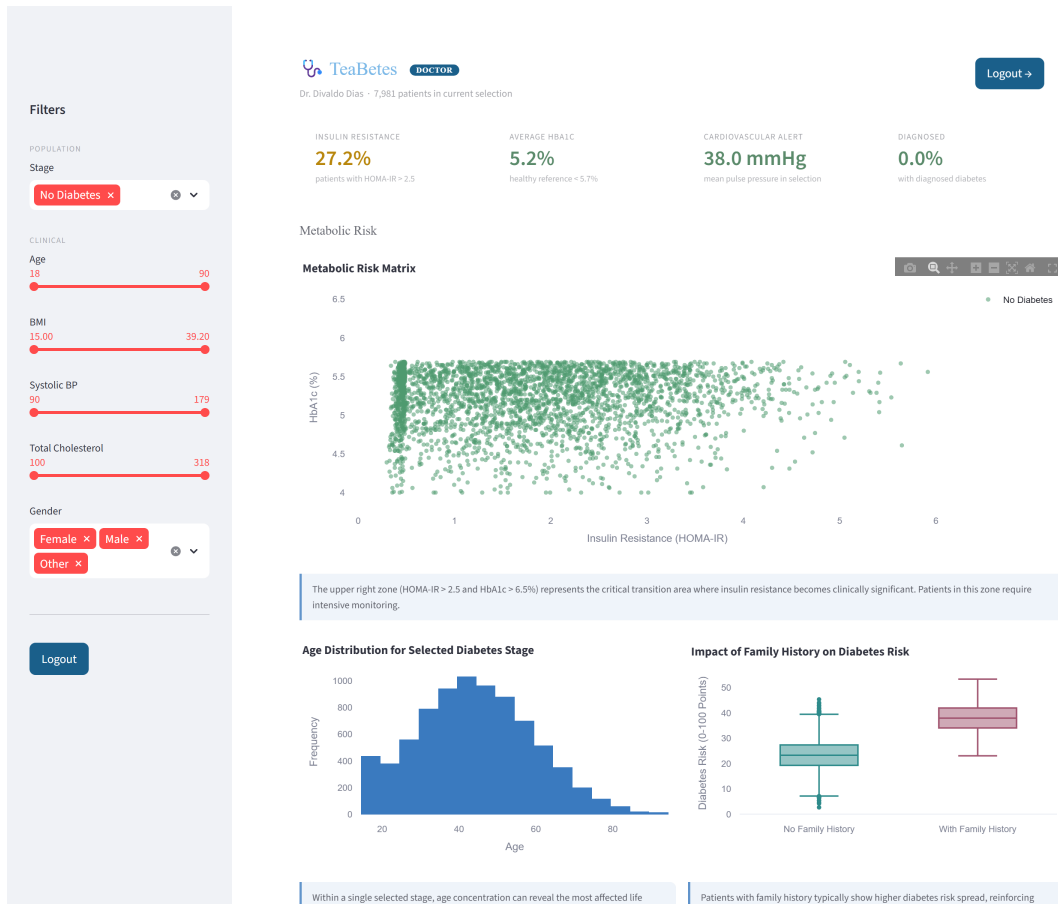


Fig. 10: Doctor dashboard overview with filtering controls, KPI cards, metabolic scatter plot, and exploratory cohort summaries.

evaluation and dashboard integration, it may not fully represent real-world clinical risk estimation scenarios.

Third, the clustering analysis should be interpreted as exploratory rather than clinically validated patient stratification. Although the elbow method and hierarchical clustering suggested the existence of latent subgroup structures, no external clinical validation, stability assessment, or longitudinal evaluation was performed.

A limitation of the deployed Stacking Regressor is its reduced ability to extrapolate beyond the clinical ranges represented in the training data. Since the ensemble is primarily composed of tree-based learners combined through a RidgeCV meta-model, predictions are naturally constrained by the patterns observed during training. Consequently, risk estimates for highly atypical patient profiles should be interpreted cautiously.

Furthermore, although the classification and regression pipelines capture complementary aspects of patient health and were successfully integrated into the deployed dashboard, their predictive performance remains dependent on the characteristics, assumptions, and representativeness of the original dataset. Therefore, the generated predictions should be interpreted as decision-support indicators rather than definitive clinical recommendations.

Finally, although multiple machine learning models and optimization strategies were evaluated, the study does not include statistical significance testing, repeated cross-validation experiments, external validation datasets, or prospective clinical evaluation. These limitations may affect the generalizability of the reported results and should be addressed in future research.

V. CONCLUSION

This work presented a comprehensive healthcare analytics pipeline for diabetes-related data analysis, combining exploratory data analysis, feature engineering, supervised learning, dimensionality reduction, clustering techniques, and interactive visualization systems.

The experimental results demonstrated that different machine learning models exhibit distinct strengths depending on the evaluation objective. Recall-oriented approaches such as the Bagging Classifier and Logistic Regression achieved the strongest sensitivity performance, making them particularly suitable for healthcare screening scenarios where minimizing false negatives is clinically important. In contrast, ensemble methods such as AdaBoost, Stacking, and XGBoost achieved the best overall balance between predictive performance and computational efficiency.

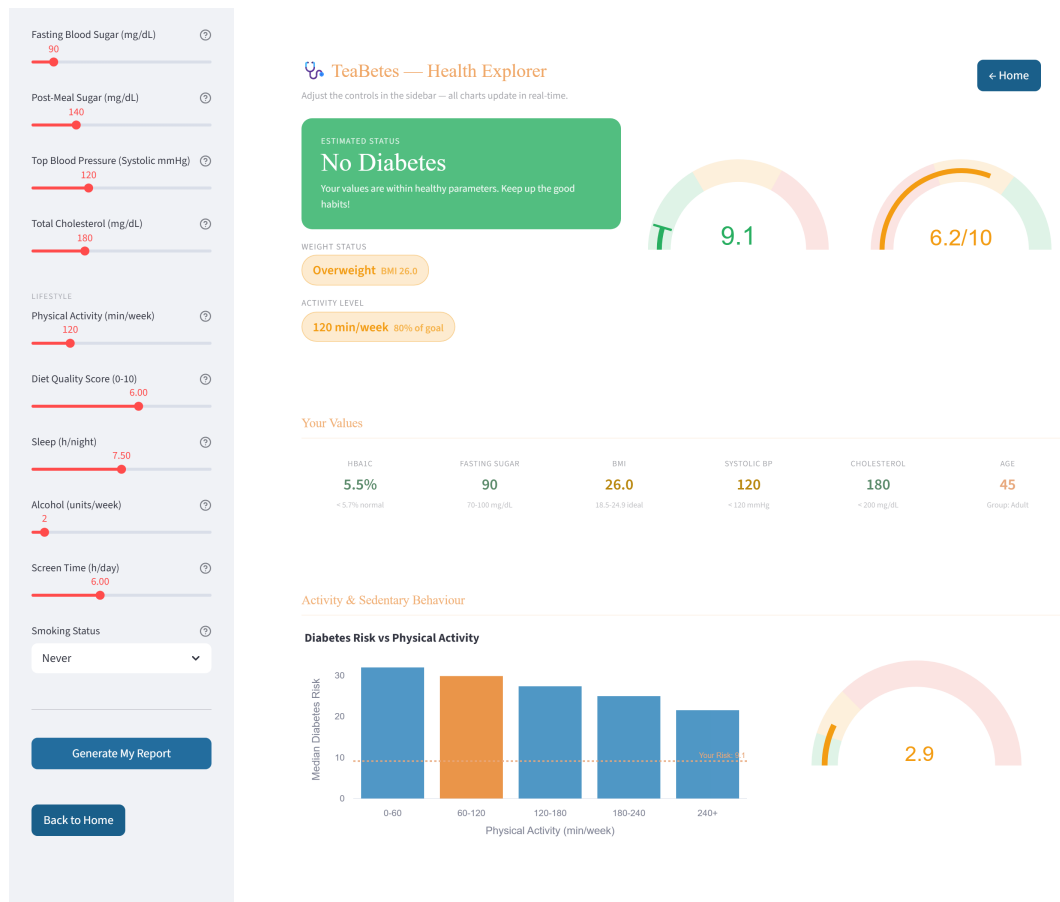


Fig. 11: Patient dashboard overview with machine-learning-based diabetes diagnosis prediction, continuous risk score estimation, lifestyle indicators, clinical value cards, and activity-risk visualization.

The regression experiments revealed extremely strong predictive performance across several models. The deployed Stacking Regressor achieved the best overall performance among the models evaluated on the complete dataset, demonstrating excellent reconstruction capability while providing robust and stable behavior suitable for real-time deployment. These results indicate that the `diabetes_risk_score` target is strongly associated with the available clinical and behavioral variables, characterizing the regression task as a clinical risk reconstruction problem rather than a purely latent prediction problem.

The dimensionality reduction experiments showed that PCA can provide important benefits for clustering and exploratory analysis while negatively impacting tree-based regression models. This finding reinforces that dimensionality reduction techniques must be evaluated according to the characteristics of each algorithm family and analytical objective.

The clustering analysis suggested the existence of potential exploratory subgroup structures within the dataset, while the developed Streamlit platform successfully integrated predictive inference and healthcare analytics into a unified decision-support environment. Through the deployment of the Bagging Classifier and Stacking Regressor, the platform supports both

real-time patient-level risk assessment and population-level exploratory analysis.

Overall, the proposed methodology demonstrated that combining feature engineering, machine learning, exploratory analytics, and interactive visualization can produce robust and interpretable healthcare analytics systems for diabetes-related studies.

As part of future work, the application will be expanded to include an AI-driven Clinical Recommendation System. While the current dashboard provides basic lifestyle-oriented feedback, the next iteration will incorporate advanced recommender-system techniques, including collaborative filtering and content-based filtering approaches inspired by personalized medicine. By clustering patients with similar metabolic profiles and analyzing the historical effectiveness of specific interventions, the system will dynamically generate tailored dietary, physical activity, and behavioral recommendations, enhancing the platform's utility as a comprehensive healthcare intervention tool. Additional research may also investigate model explainability techniques, longitudinal patient monitoring, external validation using real-world healthcare datasets, and integration with electronic health record systems.

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